

RRB JE CBT-1

Complete Study Material with Solved Examples

Section	Questions	Marks
Mathematics	30	30
General Intelligence & Reasoning	25	25
General Awareness	15	15
General Science	30	30
Total	100	100

Duration: 90 minutes (120 minutes for eligible PwD candidates) | Marking: +1 correct, -1/3 wrong | Screening stage (marks not added to final merit)

Covers: Mathematics • General Intelligence & Reasoning • General Awareness • General Science with topic-wise theory, formulas, and fully solved examples for every section.

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1. Exam Pattern & Strategy

The RRB JE CBT-1 is a common screening test for all engineering branches (Mechanical, Electrical, Civil, and others). It does not include branch-specific technical questions — those appear only in CBT-2. CBT-1 marks are used only for shortlisting; they are not carried forward to the final merit list.

Key Facts

- Total Questions: 100 | Total Marks: 100 | Duration: 90 minutes (120 minutes for eligible PwD candidates)
- Negative marking: $-1/3$ mark for every wrong answer
- All four sections (Maths, Reasoning, GA, Science) are compulsory
- Qualifying marks vary by category: UR/EWS 40%, OBC-NCL/SC 30%, ST 25% (subject to official notification)

Suggested Time Allocation (out of 90 minutes)

Section	Questions	Suggested Time
Mathematics	30	~30 min
General Intelligence & Reasoning	25	~22 min
General Awareness	15	~10 min
General Science	30	~20 min
Buffer / Revision	-	~8 min

2. Mathematics (30 Questions)

2.1 Number System, HCF & LCM

- HCF (Highest Common Factor) — largest number dividing all given numbers exactly
- LCM (Lowest Common Multiple) — smallest number divisible by all given numbers
- Key relation: $\text{HCF} \times \text{LCM} = \text{Product of two numbers}$

Example: HCF & LCM

Find the HCF and LCM of 24 and 36.

$$24 = 2^2 \times 3, \quad 36 = 2^2 \times 3^2$$

$$\text{HCF} = 2^2 \times 3 = 12$$

$$\text{LCM} = 2^2 \times 3^2 = 36 \times 24 / 12 = 72$$

Answer: HCF = 12, LCM = 72

2.2 Percentage

- $x\%$ of a number $N = (x/100) \times N$
- To find what percentage A is of B : $(A/B) \times 100$

Example: Percentage

In an exam, a student scored 462 marks out of 700. What percentage did the student get?

$$\text{Percentage} = (462 / 700) \times 100$$

$$= 46200 / 700 = 66\%$$

Answer: 66%

2.3 Profit & Loss

- Profit = Selling Price (SP) – Cost Price (CP)
- Profit % = $(\text{Profit} / \text{CP}) \times 100$
- Loss % = $(\text{Loss} / \text{CP}) \times 100$

Example: Profit & Loss

A shopkeeper bought an article for ₹800 and sold it for ₹920. Find the profit percentage.

$$\text{Profit} = 920 - 800 = ₹120$$

$$\text{Profit \%} = (120 / 800) \times 100 = 15\%$$

Answer: 15% profit

2.4 Simple Interest & Compound Interest

- Simple Interest (SI) = $(P \times R \times T) / 100$
- Compound Interest (CI) = $P(1 + R/100)^T - P$

Example: Simple Interest

Find the SI on ₹5,000 at 8% per annum for 3 years.

$$\text{SI} = (5000 \times 8 \times 3) / 100$$

$$= 120000 / 100 = ₹1,200$$

Answer: ₹1,200

2.5 Time, Speed & Distance

- Speed = Distance / Time
- 1 km/h = $\frac{5}{18}$ m/s, and 1 m/s = $\frac{18}{5}$ km/h
- For trains crossing objects, relative speed rules apply for same/opposite direction motion

Example: Speed, Time & Distance

A train 150 m long crosses a pole in 10 seconds. Find its speed in km/h.

Speed = Distance / Time = $150 / 10 = 15$ m/s

Convert to km/h: $15 \times \frac{18}{5} = 54$ km/h

Answer: 54 km/h

2.6 Time & Work

- If A can do a work in n days, A's 1-day work = $\frac{1}{n}$
- Combined work rate = sum of individual work rates

Example: Time & Work

A can complete a work in 12 days and B can complete it in 18 days. In how many days will they finish it together?

A's 1-day work = $\frac{1}{12}$, B's 1-day work = $\frac{1}{18}$

Combined 1-day work = $\frac{1}{12} + \frac{1}{18} = \frac{(3+2)}{36} = \frac{5}{36}$

Time taken together = $\frac{36}{5} = 7.2$ days

Answer: 7.2 days

2.7 Average

Example: Average

The average of 5 numbers is 24. If one number is excluded, the average becomes 22. Find the excluded number.

Sum of 5 numbers = $24 \times 5 = 120$

Sum of remaining 4 numbers = $22 \times 4 = 88$

Excluded number = $120 - 88 = 32$

Answer: 32

2.8 Algebra

- $(a+b)^2 = a^2 + 2ab + b^2$
- $(a-b)^2 = a^2 - 2ab + b^2$
- $a^2 - b^2 = (a+b)(a-b)$

Example: Algebra

If $x + \frac{1}{x} = 5$, find the value of $x^2 + \frac{1}{x^2}$.

$x^2 + \frac{1}{x^2} = (x + \frac{1}{x})^2 - 2$

$= 5^2 - 2 = 25 - 2 = 23$

Answer: 23

2.9 Mensuration

- Area of rectangle = length $\times$ breadth; Area of triangle = $\frac{1}{2} \times$ base $\times$ height
- Area of circle = πr^2 ; Circumference = $2\pi r$
- Volume of cylinder = $\pi r^2 h$; Volume of cube = a^3

Example: Mensuration

Find the area of a circle with radius 7 cm. (Use $\pi = 22/7$)

$$\text{Area} = \pi r^2 = 22/7 \times 7 \times 7$$

$$= 22 \times 7 = 154 \text{ cm}^2$$

Answer: 154 cm²

2.10 Trigonometry (Heights & Distances)

- $\sin\theta = \text{Opposite/Hypotenuse}$, $\cos\theta = \text{Adjacent/Hypotenuse}$, $\tan\theta = \text{Opposite/Adjacent}$
- Standard values: $\sin 30^\circ = 1/2$, $\cos 30^\circ = \sqrt{3}/2$, $\tan 45^\circ = 1$

Example: Heights & Distances

The angle of elevation of the top of a tower from a point 50 m away is 30° . Find the height of the tower.

$$\tan 30^\circ = \text{Height} / 50$$

$$1/\sqrt{3} = \text{Height} / 50$$

$$\text{Height} = 50/\sqrt{3} \approx 28.87 \text{ m}$$

Answer: approx. 28.87 m

3. General Intelligence & Reasoning (25 Questions)

3.1 Number & Alphabet Series

Example: Number Series

Find the next number in the series: 3, 7, 15, 31, 63, ?

Pattern: each term = previous term $\times 2 + 1$

$3 \times 2 + 1 = 7$, $7 \times 2 + 1 = 15$, $15 \times 2 + 1 = 31$, $31 \times 2 + 1 = 63$

Next: $63 \times 2 + 1 = 127$

Answer: 127

3.2 Coding-Decoding

Example: Coding-Decoding

If CAT is coded as DBU, how is DOG coded in the same pattern?

Each letter is shifted forward by 1: C $\rightarrow$ D, A $\rightarrow$ B, T $\rightarrow$ U

Apply same rule to DOG: D $\rightarrow$ E, O $\rightarrow$ P, G $\rightarrow$ H

Answer: EPH

3.3 Analogy

Example: Analogy

Book is to Reading as Fork is to ?

A Book is the tool/medium used for the action of Reading.

Similarly, a Fork is the tool used for the action of Eating.

Answer: Eating

3.4 Blood Relations

Example: Blood Relations

Pointing to a man, a woman said, "His mother is the only daughter of my mother." How is the woman related to the man?

"Only daughter of my mother" = the woman herself (she is her mother's only daughter)

So "his mother" = the woman herself

That means the man is the woman's son

Answer: Mother

3.5 Syllogism

Example: Syllogism

Statements: All pens are pencils. All pencils are erasers.

Conclusion I: All pens are erasers.

Conclusion II: Some erasers are pens.

All pens $\subseteq$ pencils, and all pencils $\subseteq$ erasers $\Rightarrow$ all pens $\subseteq$ erasers (Conclusion I follows)

Since all pens are erasers, at least some erasers must be pens (Conclusion II follows)

Answer: Both conclusions I and II follow

3.6 Direction Sense

Example: Direction Sense

Ravi walks 5 km North, then turns right and walks 3 km, then turns right again and walks 5 km.

How far is he from the starting point?

North 5 km, then East 3 km (turned right from North), then South 5 km (turned right from East)

Net displacement: North and South cancel out (5 km each), only the 3 km East move remains

Answer: 3 km (to the East of the starting point)

3.7 Classification (Odd One Out)

Example: Classification

Find the odd one out: Triangle, Square, Circle, Pentagon

Triangle, Square, and Pentagon are all polygons with straight sides.

Circle is the only figure with no straight sides/vertices.

Answer: Circle

Other topics to practice: mathematical operations (symbol-based), sentence/word jumbling, Venn diagrams, data sufficiency, statement-conclusion, statement-assumption, similarities & differences, map-based reasoning.

4. General Awareness (15 Questions)

This section covers static GK (polity, geography, history, economy) and current affairs (national & international, sports, science & tech). Below are representative practice questions in each area — use them as a model for the type of one-liner facts to revise, and keep a running current-affairs file for the 3–4 months before your exam.

4.1 Indian Polity

Example: Polity

Q. How many fundamental rights are guaranteed by the Indian Constitution?

A. Six fundamental rights (originally seven; the Right to Property was removed as a fundamental right by the 44th Amendment).

4.2 Geography

Example: Geography

Q. Which is the longest river in India?

A. The Ganga (Ganges), flowing through northern India.

4.3 History

Example: History

Q. Who was the first Governor-General of independent India?

A. Lord Mountbatten (1947–1948), followed by C. Rajagopalachari as the first and only Indian Governor-General.

4.4 Economy

Example: Economy

Q. Which organisation regulates monetary policy in India?

A. The Reserve Bank of India (RBI), India's central bank.

4.5 Sports & Current Affairs

- Track recent RRB/Railway-related news, government schemes, national awards, and sporting events (Olympics, Asian Games, major tournaments)
- Note important days & their significance (e.g., National Science Day, World Environment Day)
- Revise recently released books, appointments (RBI Governor, Chief Justice, Army/Navy/Air Force chiefs), and index rankings (India's rank in global reports)

Static GK topics to cover thoroughly: Indian National Movement, Constitution & governance, physical & economic geography of India, world geography basics, first-in-India facts, important rivers/dams/national parks, awards & honours.

5. General Science (30 Questions) — up to Class 10 level

5.1 Physics

- Laws of Motion (Newton's three laws), work-energy-power, gravitation, sound & light (reflection/refraction), heat & thermodynamics basics, electricity & magnetism basics, simple machines

Example: Physics

Q. A body of mass 2 kg is moving with a velocity of 10 m/s. Find its kinetic energy.

$$KE = \frac{1}{2}mv^2 = \frac{1}{2} \times 2 \times 10^2$$

$$= \frac{1}{2} \times 2 \times 100 = 100 \text{ J}$$

Answer: 100 Joules

5.2 Chemistry

- States & properties of matter, atomic structure, periodic table trends, chemical reactions & balancing equations, acids/bases/salts, metals & non-metals, basic carbon compounds

Example: Chemistry

Q. What is the pH value of a neutral solution?

A. 7 (pH below 7 is acidic, above 7 is basic/alkaline).

5.3 Biology

- Cell structure & function, human body systems (digestive, respiratory, circulatory, nervous), nutrition & diseases, plant physiology basics, basic genetics

Example: Biology

Q. Which is the largest organ in the human body?

A. The Skin.

5.4 Quick Fact Table — Frequently Asked Science Points

Topic	Key Fact
SI unit of Force	Newton (N)
SI unit of Power	Watt (W)
Gas responsible for greenhouse effect	Carbon dioxide (CO ₂)
Normal human body temperature	37°C (98.6°F)
Chemical formula of table salt	NaCl (Sodium Chloride)
Powerhouse of the cell	Mitochondria
Speed of light in vacuum	3×10^8 m/s

NCERT Class 9 & 10 Science textbooks cover almost all General Science topics tested in CBT-1 — treat them as the primary source, then revise using a compact GK book for quick recall.

6. Recommended Study Material

Mathematics

- Fast Track Objective Arithmetic — Rajesh Verma
- Quantitative Aptitude for Competitive Examinations — R.S. Aggarwal
- Practice Book on Quicker Maths — M. Tyra / K. Kundan
- NCERT Mathematics, Class 9–10 (for concept clarity on geometry, mensuration, algebra)

General Intelligence & Reasoning

- A Modern Approach to Verbal & Non-Verbal Reasoning — R.S. Aggarwal
- Test of Reasoning — Edgar Thorpe
- General Intelligence and Reasoning — Kunwar Ishwar Singh Rathore

General Awareness

- Lucent's General Knowledge — Lucent Publication (most recommended single GK book for RRB exams)
- Pratiyogita Darpan (monthly magazine, for current affairs)
- Manorama Yearbook — for a consolidated annual GK reference

General Science

- NCERT Physics, Chemistry & Biology — Class 9 & 10 (core, high-yield source)
- Lucent's General Science — as a compact revision book

Practice & Mocks

- RRB JE CBT-1 previous-year solved papers (Oswaal / Disha / Mocktime publications have chapter-wise PYQ books)
- A full-length CBT-1 mock test series (Made Easy, Physics Wallah, Testbook, etc.) to build 90-minute exam-temperament

How to use these: Treat NCERT (Maths + Science) as your concept foundation, the subject-specific books above for practice volume, and PYQ/mock-test books for exam-pattern familiarity in the final month.

7. Final Preparation Tips

- Give equal weight to Mathematics and Reasoning early — together they're 55/100 marks and most scoring with practice
- Revise NCERT Class 9–10 Science thoroughly — factual, high-yield, and repeats across exam years
- Spend ~30 minutes daily on current affairs; do one full static-GK revision cycle before the exam
- Attempt full-length CBT-1 mocks weekly under strict 90-minute timing once basics are covered
- Because of $-1/3$ negative marking, avoid guessing unless you can confidently eliminate at least 2 of the 4 options
- Maintain a formula sheet (Maths) and a shortcut/trick sheet (Reasoning) for quick revision in the last week

Note: The official CBT-1 syllabus and pattern are notified separately for each RRB JE recruitment cycle (CEN). Always cross-check against the latest official notification on your regional RRB website before finalizing your prep plan.